

Question 1

6 parts | – of 1 point

Arbor Systems and Gencore stocks both have a volatility of 43%. Compute the volatility of a portfolio with 50% invested in each stock if the correlation between the stocks is **(a)** +1.00, **(b)** 0.50, **(c)** 0.00, **(d)** –0.50, and **(e)** –1.00. In which of the cases is the volatility lower than that of the original stocks?

Review Only

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Part 1 of 6

If the correlation is +1.00, the volatility of the portfolio is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 2 of 6

If the correlation is 0.50, the volatility of the portfolio is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 3 of 6

If the correlation is 0.00, the volatility of the portfolio is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 4 of 6

If the correlation is -0.50 , the volatility of the portfolio is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 5 of 6

If the correlation is -1.00 , the volatility of the portfolio is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 6 of 6

In which of the cases is the volatility lower than that of the original stocks? (Select the best choice.)

- ☐ A. In all of the cases.
- ☐ B. In cases **(b)**, **(c)**, **(d)** and **(e)**.
- ☐ C. In cases **(d)** and **(e)**.
- ☐ D. In none of the cases.

14 parts | – of 1 point

- What is the return each year of this portfolio?
- Based on your results from part **(a)**, compute the average return and volatility of the portfolio.
- Show that (i) the average return of the portfolio is equal to the (weighted) average of the average returns of the two stocks, and (ii) the volatility of the portfolio equals the same result as from the calculation in Eq. 11.8.
- Explain why the portfolio has a lower volatility than the average volatility of the two stocks.

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Year	2010	2011	2012	2013	2014	2015
Stock A	−8%	12%	10%	−8%	5%	6%
Stock B	19%	16%	4%	−7%	−6%	28%

[illegible]

Part 2 of 13

b. Based on your results from part **(a)**, compute the average return and volatility of the portfolio.

The average return of the portfolio is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 3 of 13

The variance of the portfolio is (Round your answer to five decimal places.) .

Part 4 of 13

The standard deviation of the portfolio is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 5 of 13

c. Show that (i) the average return of the portfolio is equal to the (weighted) average of the average returns of the two stocks, and (ii) the volatility of the portfolio equals the same result as from the calculation in Eq. 11.8.

The average annual return for stock A is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 6 of 13

The average annual return for stock B is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 7 of 13

The (weighted) average of the average returns of the two stocks is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 8 of 13

The standard deviation of stock A is (Round your answer to five decimal places.) .

Part 9 of 13

The standard deviation of stock B is (Round your answer to five decimal places.) .

Part 10 of 13

The covariance between the stocks is (Round your answer to five decimal places.) .

Part 11 of 13

The variance of the portfolio is (Round your answer to five decimal places.) .

Part 12 of 13

The volatility of the portfolio is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 13 of 13

d. Explain why the portfolio has a lower volatility than the average volatility of the two stocks.

The portfolio has a (Select your answer from the drop-down menu.) **Select ▼** volatility than the average volatility of the two stocks because some of the idiosyncratic risk of the stocks in the portfolio is diversified away.

Question 3

9 parts | – of 1 point

You own three stocks: 600 shares of Apple Computer, 10,000 shares of Cisco System, and 5,000 shares of Colgate-Palmolive. The current share prices and expected returns of Apple, Cisco, and Colgate-Palmolive are, respectively, \$482, \$16, \$101 and 12%, 10%, 8%.

- What are the portfolio weights of the three stocks in your portfolio?
- What is the expected return of your portfolio?
- Suppose the price of Apple stock goes up by \$28, Cisco rises by \$4, and Colgate-Palmolive falls by \$15. What are the new portfolio weights?
- Assuming the stocks' expected returns remain the same, what is the expected return of the portfolio at the new prices?

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Part 1 of 8

- a. What are the portfolio weights of the three stocks in your portfolio?

The portfolio weight of Apple is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 2 of 8

The portfolio weight of Cisco is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 3 of 8

The portfolio weight of Colgate-Palmolive is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 4 of 8

b. What is the expected return of your portfolio?

The expected return on the portfolio is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 5 of 8

c. Suppose the price of Apple stock goes up by \$28, Cisco rises by \$4, and Colgate-Palmolive falls by \$15. What are the new portfolio weights?

The new portfolio weight of Apple is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 6 of 8

The new portfolio weight of Cisco is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 7 of 8

The new portfolio weight of Colgate-Palmolive is (Enter your answer as a percentage and round it to two decimal places.) %.

Part 8 of 8

d. Assuming the stocks' expected returns remain the same, what is the expected return of the portfolio at the new prices?

The new expected return is (Enter your answer as a percentage and round it to two decimal places.) %.

Question 4

3 parts | – of 1 point

Suppose Johnson & Johnson and Walgreens Boots Alliance have expected returns and volatilities shown [here](#) with a correlation of 16%. Calculate **(a)** the expected return and **(b)** the volatility (standard deviation) of a portfolio that consists of a long position of \$9,500 in Johnson & Johnson and a short position of \$2,000 in Walgreens.

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Part 1 of 2

a. Calculate the expected return of a portfolio that consists of a long position of \$9,500 in Johnson & Johnson and a short position of \$2,000 in Walgreens.

The expected return of the portfolio is (Enter your answer as a percentage and round it to two decimal place.) %.

Part 2 of 2

b. Calculate the volatility (standard deviation) of a portfolio that consists of a long position of \$9,500 in Johnson & Johnson and a short position of \$2,000 in Walgreens.

The volatility of the portfolio is (Enter your answer as a percentage and round it to two decimal place.) %.

Question 5

4 parts | – of 1 point

Suppose Intel's stock has an expected return of 22% and a volatility of 32%, while Coca-Cola's has an expected return of 9% and volatility of 19%. If these two stocks were perfectly negatively correlated (i.e., their correlation coefficient is -1),

- Calculate the portfolio weights that remove all risk.
- If there are no arbitrage opportunities, what is the risk-free rate of interest in this economy?

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Part 1 of 3

- a. Calculate the portfolio weights that remove all risk.

The portfolio weight of Intel would be (Enter your answer as a percentage and round it to two decimal places.) %.

Part 2 of 3

The portfolio weight of Coca-Cola would be (Enter your answer as a percentage and round it to two decimal places.) %.

Part 3 of 3

- b. If there are no arbitrage opportunities, what is the risk-free rate of interest in this economy?

The risk-free rate of interest in this economy is (Enter your answer as a percentage and round it to two decimal places.) %.